



Dharussalam Primary School

Thihariya · WP / GM · established 2012

Software Requirements Specification · Version 1.0 (draft)

School Administration Mobile Application

Requirements for the staff-facing mobile application supporting Dharussalam Primary School, Thihariya — approximately 900 students and 35 staff.

School	Document	Status	Scope	Companion
Dharussalam Primary School	SRS — Admin App	Draft for review	Phase 1 (admin app)	AI Build Specification

Contents

1	Purpose and scope
2	Glossary
3	System overview
4	Roles and permissions
5	Functional requirements
6	Screens and wireframes
7	Data model
8	Non-functional requirements
9	Delivery phases
10	Assumptions and exclusions
11	Open decisions

1 Purpose and scope

This document specifies what the school administration mobile application must do and how well it must do it. It is written to be read and approved by Dharussalam Primary School before development begins, and to be referred to during development as the single description of intended behaviour.

1.1 Background

Dharussalam Primary School currently keeps its records in spreadsheets. Attendance is taken on paper and transcribed. There is no reliable way for the principal to know, at any moment in the morning, whether every class has been marked, and no systematic way to notice that a child has stopped coming to school until the absence is already long-standing.

1.2 Scope of this release

This specification covers the **admin application** only — the app used by the principal, sectional heads, class teachers, administrators and support staff. A separate parent-facing application is planned for a later phase and is described here only where phase 1 must lay groundwork for it (for example, guardian records and contact numbers).

1.3 Objectives

- Attendance for every class is recorded before the school day begins, from a phone, in under thirty seconds.
- The principal can see the state of the whole school on one screen and act on what is wrong without leaving it.
- A child whose attendance is deteriorating is surfaced automatically rather than noticed by chance.
- Staff records, duties, leave and school assets are held in one place instead of several spreadsheets.
- All school data is stored in a proper database with per-role access control and an audit trail.

1.4 Success measures

- Over 95% of classes marked by the daily attendance deadline within one term of launch.
- Zero paper attendance registers in use by the end of term following launch.
- Every absence of three or more consecutive days followed up and the follow-up recorded.

2 Glossary

Term	Meaning
Class	A single teaching group, e.g. 4B. Dharussalam Primary School has roughly 30 classes across grades 1–5.
Section	All classes of one grade, e.g. grade 4 = 4A–4E. Overseen by a sectional head.
Scope	The set of records a role may act on: school-wide, one section, or one class.
School day	A day on which attendance is expected. Derived from term dates, the working-day pattern and any day-level override.
Attendance deadline	The time by which each class must have submitted attendance. Unmarked classes are escalated after it.
Edit window	The period after submission during which a teacher may correct attendance without an administrator.
Cover teacher	A teacher temporarily granted a class teacher's rights over a class while its own teacher is absent.
Guardian	A parent or other adult responsible for a student. A guardian may be linked to several students.
Benefit	A support scheme a student receives — free textbooks, uniform voucher, midday meal, season ticket.

3 System overview

The system is a mobile application backed by a hosted relational database and API. The application is built once and runs on both Android and iOS. All business rules, permission checks and scheduled jobs live on the server; the app is a client.

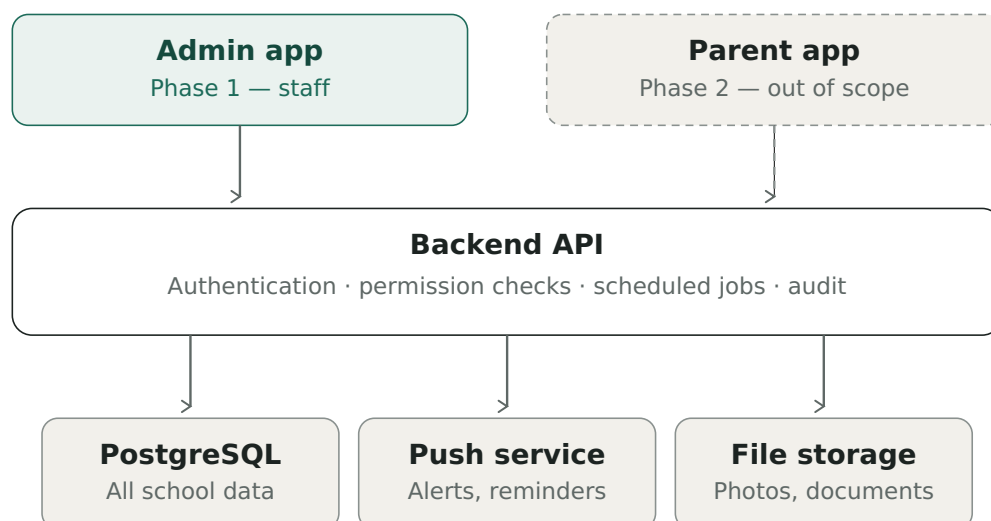


Figure 1. System architecture. The parent application is shown for context only; it is not part of this release.

3.1 Technology

Layer	Choice	Reason
Mobile client	React Native	One codebase for Android and iOS; existing team familiarity.
Database	PostgreSQL	Relational data with concurrent writers; spreadsheets cannot support 30 teachers marking at once.
Backend platform	Managed Postgres platform providing authentication, row-level security, storage and scheduled functions	Removes the need to build and operate authentication and file storage separately.
Notifications	Platform push service	Required for attendance reminders and absence alerts.
Existing data	One-time import from spreadsheets	Spreadsheets are the source for initial student and staff records only. They are not a live backend.

4 Roles and permissions

Roles are not fixed in code. A role is a named set of permissions, and a person is assigned a role together with a **scope** — the school, one section, or one class. A person may hold more than one role at once; a sectional head is normally also a class teacher, and the system must serve both roles in one session without switching profiles.

4.1 Roles

Role	Scope	Responsibility
Principal	School	Full read across the school; approves leave; broadcasts announcements; administers the calendar, accounts and inventory.
Vice-principal	School	As principal, with leave approval optional per school policy.
Sectional head	Section	Oversees the classes of one grade; assigns cover teachers; reviews section attendance and marks.
Class teacher	Class	Marks attendance, records early leaves, enters marks, maintains records for their own class.
Administrator	School	Manages user accounts, student and staff records, inventory and benefits. No access to marks or analytics.
Staff	Self	Own profile, own attendance, own leave, staff directory, announcements.

4.2 Permission matrix

Capability	Prin.	Sect.	Class	Admin	Staff
Mark student attendance	All	Section	Own	—	—
View attendance board	All	Section	Own	—	—
Send unmarked-class reminder	All	Section	—	—	—
Record early leave	All	Section	Own	—	—
Enter marks	All	Section	Own	—	—
View student profile (full)	All	Section	Own	All	—
View student benefits	Yes	—	Own	Yes	—
View guardian contact numbers	All	Section	Own	All	—
Approve leave	Yes	—	—	—	—
Request leave	Yes	Yes	Yes	Yes	Yes
Assign cover teacher	All	Section	—	—	—
Assign responsibilities	Yes	Section	—	—	—
Publish announcement	Any	Section	—	Staff	—
Manage inventory	Yes	—	—	Yes	—
Manage events and diary	Yes	—	—	Yes	—
Edit academic calendar	Yes	—	—	TBD	—
Manage user accounts and roles	Yes	—	—	Yes	—
View analytics	All	Section	Own	—	—
View audit log	Yes	—	—	—	—
Staff directory and call	Yes	Yes	Yes	Yes	Yes

Permission model

Every permission check answers one question: does any role assignment held by this user grant this permission at a scope that contains this record? Adding a role, or a deputy for grades 1–2, is then a data change rather than a code change.

5 Functional requirements

Requirements are grouped by module and carry a stable identifier. Priority **P1** must be present at launch, **P2** follows within the first term, **P3** is desirable.

5.1 Authentication and accounts AUTH

FR-AUTH-01 A staff member signs in with a staff identifier and password. Accounts are created by an administrator; there is no self-registration. **P1**

FR-AUTH-02	On first sign-in the user must set a new password before reaching any other screen. P1
FR-AUTH-03	A user may reset a forgotten password by one-time code sent to their registered mobile number. P1
FR-AUTH-04	After first sign-in the user may enable device biometric unlock for subsequent sessions. P2
FR-AUTH-05	A session remains valid for a configurable period and renews on use, so a teacher is not asked to sign in every morning. P1
FR-AUTH-06	A deactivated account cannot sign in and its existing sessions end within one minute. P1
FR-AUTH-07	The interface presented after sign-in is determined by the user's role assignments. No role selection screen is shown. P1

5.2 Academic calendar CAL

FR-CAL-01	The principal defines an academic year with a start and end date. P1
FR-CAL-02	The principal defines terms within the year, each with a start and end date. Terms may not overlap. P1
FR-CAL-03	The principal sets the weekly working-day pattern for the school. P1
FR-CAL-04	A day is a school day if it falls inside a term, on a working weekday, and carries no override to the contrary. School days are derived, not entered one by one. P1
FR-CAL-05	The principal may override any individual day as holiday, half day or examination day, with an optional label. P1
FR-CAL-06	The principal may declare an emergency closure for the current or a future day from the home screen. This cancels that day's attendance obligations, removes the day from attendance calculations, and notifies all staff. P1 Rationale: a closed day must not be recorded as 900 absences.
FR-CAL-07	Changing the type of a past day recalculates affected attendance percentages, and the user is warned before saving. P2
FR-CAL-08	The principal sets the daily attendance deadline and the length of the correction window after submission. P1
FR-CAL-09	Public holidays may be imported in bulk for a year rather than entered individually. P3

5.3 Students and guardians STU

FR-STU-01	Each student holds an admission number unique within the school, full name, date of birth, current class, and optional photograph. P1
-----------	--

FR-STU-02	A student may be linked to one or more guardians, and a guardian may be linked to more than one student, so that siblings resolve to the same contact. P1
FR-STU-03	Exactly one guardian per student is marked as the primary contact. P1
FR-STU-04	Students may be searched by name or admission number, including partial admission numbers, within the user's scope. P1
FR-STU-05	Students may be browsed by grade and class. P1
FR-STU-06	The student profile presents overview, attendance, marks, activity and benefits. Tabs the user may not see are not displayed. P1
FR-STU-07	A permitted user may place a telephone call to a guardian directly from the student record. P1
FR-STU-08	After a call placed from the app, the user is prompted to record its purpose in one line, retained against the student. P2
FR-STU-09	The student overview shows one chronological timeline combining absences, early leaves, logged calls, achievements and benefit changes. P2
FR-STU-10	A user viewing a student outside their scope sees name, class and photograph only. P1
FR-STU-11	Students may be promoted in bulk to the next grade at the start of an academic year, retaining their history. P2
FR-STU-12	A student may be marked as left, retaining all records but excluded from rosters and counts. P1

5.4 Student attendance ATT

FR-ATT-01	The marking roster loads with every student defaulted to present. The teacher marks only the exceptions. P1 Rationale: the single largest factor in daily adoption.
FR-ATT-02	A student may be marked present, absent or late. P1
FR-ATT-03	Students absent on the previous school day are visually indicated on the roster while marking. P2
FR-ATT-04	Attendance may be marked without network connectivity. Entries are held on the device and transmitted when connectivity returns, and the app shows clearly whether the record has reached the server. P1
FR-ATT-05	Attendance is submitted once per class per school day. A later submission replaces the first only within the correction window. P1

FR-ATT-06	After the correction window closes, only the principal or an administrator may amend that day's attendance, and every amendment is recorded in the audit log with a reason. P1
FR-ATT-07	On submission the teacher is shown the absentees, each with the number of consecutive days absent and a means of calling the guardian. P1
FR-ATT-08	A reason may be attached to an absence at any later time. P2
FR-ATT-09	A teacher may record a student's early departure with time, reason and the person collecting them. P1
FR-ATT-10	The attendance board shows, for a selected date, every class in the user's scope and whether it has been marked, with unmarked classes ordered first. P1
FR-ATT-11	Selecting a class on the board shows the students absent from it that day. P1
FR-ATT-12	The principal or a sectional head may remind the teachers of all unmarked classes in their scope in one action. P1
FR-ATT-13	The system does not expect, request or alert on attendance for any day that is not a school day. P1
FR-ATT-14	Attendance history for a student is presented as a monthly calendar of school days only, distinguishing present, absent, late and early leave. P1

5.5 Marks and assessment MRK

FR-MRK-01	Subjects are defined per grade. Each class-subject pairing has an assigned teacher. P1
FR-MRK-02	A teacher enters marks for one class, subject and term on a single list, and may save an incomplete entry as a draft. P1
FR-MRK-03	Submitting a mark sheet locks it. Only the principal may reopen a locked sheet, and reopening is audited. P1
FR-MRK-04	Every stored mark records who entered it and when. P1
FR-MRK-05	A student's marks are shown alongside the class average for the same subject and term. P1 Rationale: a mark without a comparison carries no meaning.
FR-MRK-06	A student's term average and position within the class are shown. P2
FR-MRK-07	A student's average is presented across terms so that a trend is visible. P2
FR-MRK-08	Users with review permission see class and subject comparisons across their scope, including which mark sheets are outstanding. P2

5.6 Achievements, memberships and benefits ACH

FR-ACH-01	An achievement may be recorded against a student with title, date, category and level. P1
FR-ACH-02	A student may be recorded as a member of a club, team, society or house, with an optional position and the years held. P1
FR-ACH-03	A student may hold a class responsibility for a stated term. P2
FR-ACH-04	A benefit may be recorded against a student with scheme, period, and a status of issued, pending or active. P1
FR-ACH-05	Benefit records are visible only to the principal, an administrator, and the student's own class teacher. P1 Rationale: benefits disclose a family's financial circumstances.
FR-ACH-06	A pending benefit may be marked as collected, recording who did so and when. P2

5.7 Staff STF

FR-STF-01	Each staff member holds a staff number, name, contact number, joining date, and one or more role assignments with scope. P1
FR-STF-02	All staff may view the directory and place a call to any colleague from it. P1
FR-STF-03	The directory indicates each colleague's presence for the current day. P2
FR-STF-04	A responsibility — a duty, committee or post — may be assigned to a staff member for an academic year, and is visible both under that person and as a roster of duties. P1
FR-STF-05	A staff profile shows attendance summary, responsibilities and leave taken, subject to the viewer's permissions. P1
FR-STF-06	Ordinary staff see names, roles and contact numbers only. Attendance percentages, leave balances and role controls are not shown to them. P1

5.8 Staff attendance SAT

FR-SAT-01	A staff member records their arrival for the day in the app. P1
FR-SAT-02	Arrival after the configured start time is recorded as late. P2
FR-SAT-03	The staff attendance board distinguishes three states: present, on approved leave, and not checked in. Only the third is treated as requiring action. P1
FR-SAT-04	Where an absent staff member is a class teacher, the board shows the affected class and whether its attendance has been marked. P1
FR-SAT-05	A staff day report may be exported for a chosen date. P3

5.9 Leave LVE

FR-LVE-01	A staff member submits a leave request stating type, dates or half-day period, and reason, with an optional attachment. P1
FR-LVE-02	Leave entitlements are held per staff member, per type, per year, and the remaining balance is shown when requesting. P1
FR-LVE-03	A new request notifies the approver and appears in their pending list. P1
FR-LVE-04	Before approving, the approver is shown the requester's remaining balance, how many other staff are already on leave for those dates, and whether the requester's class needs a cover teacher. P1
FR-LVE-05	A request may only be approved from the detail view where that impact is displayed. P1 Rationale: approving without seeing the coverage gap is how a class ends up unattended.
FR-LVE-06	Approving leave for a class teacher requires a cover teacher to be nominated, or an explicit acknowledgement that none is needed. P1
FR-LVE-07	A decision may carry remarks, and the requester is notified of the outcome. P1
FR-LVE-08	Approved leave reduces the balance and marks the staff member as on leave for those dates. P1
FR-LVE-09	A staff member may withdraw a request that has not yet been decided. P2
FR-LVE-10	Every decision is recorded in the audit log with approver, timestamp and remarks. P1

5.10 Cover teachers COV

FR-COV-01	A principal or sectional head may assign a cover teacher to a class for a stated date range. P1
FR-COV-02	A cover teacher holds the class teacher's rights over that class for the duration of the assignment, and no longer. P1
FR-COV-03	Both the cover teacher and the class's own teacher are notified of the assignment. P2
FR-COV-04	Attendance marked by a cover teacher records who actually marked it. P1

5.11 Announcements ANN

FR-ANN-01	An announcement carries a title, body, optional attachment, priority and an audience. P1
FR-ANN-02	The audience may be all staff, a section, a class, or a named list of individuals. A publisher may only select audiences within their scope. P1

FR-ANN-03	Publishing sends a push notification to the audience and places the announcement in their feed. P1
FR-ANN-04	An announcement may be scheduled for later publication. P2
FR-ANN-05	The publisher can see who has read an announcement. P2
FR-ANN-06	Announcements are reachable from every screen by a single control that indicates unread items. P1

5.12 Inventory INV

FR-INV-01	An inventory item holds a name, category, quantity, location, condition and minimum quantity. P1
FR-INV-02	Items may be searched and filtered by category. P1
FR-INV-03	An item at or below its minimum quantity is flagged and ordered first in the list. P1
FR-INV-04	Movements — receipt, issue, return, write-off — are recorded as transactions, and quantity is derived from them rather than edited directly. P1
FR-INV-05	An item's transaction history is visible from the item. P2
FR-INV-06	Items may carry a printed code that can be scanned to open the item or conduct a stock check. P3

5.13 Events and school diary EVT · DRY

FR-EVT-01	An event holds a title, date or date range, description, category, location and responsible staff. P1
FR-EVT-02	Events are presented as an annual calendar in month and list views. P1
FR-EVT-03	Each event may carry one or more reminders that notify its audience a stated period beforehand. P1
FR-EVT-04	A completed event may be written into the school diary in one action, carrying its details forward. P2
FR-DRY-01	A diary entry holds a date, title, body, author and optional photographs. P1
FR-DRY-02	Diary entries are presented as a chronological record filterable by year, forming the school's continuous history. P1
FR-DRY-03	Entries not originating from an event may be added directly. P1

5.14 Notifications NOT

FR-NOT-01	At the attendance deadline on each school day, the teacher of every unmarked class is notified, and those classes appear on the principal's home screen. P1
FR-NOT-02	When a student reaches a configured number of consecutive absences, or falls below a configured attendance percentage, the class teacher, sectional head and principal are notified. P1
FR-NOT-03	Leave requests, decisions and cover assignments generate notifications to the parties concerned. P1
FR-NOT-04	Event reminders are sent according to each event's configured lead time. P1
FR-NOT-05	No scheduled notification is generated on a day that is not a school day. P1
FR-NOT-06	A user may control which categories of notification they receive, except those classed as mandatory. P2

5.15 Analytics and reports ANL

FR-ANL-01	Attendance rates are presented by class, grade and school over a chosen period, within the user's scope. P2
FR-ANL-02	Students at risk by consecutive absence or low attendance are listed, most severe first. P1
FR-ANL-03	Academic performance is presented by subject and class for a chosen term. P2
FR-ANL-04	Attendance and marks reports may be exported in a spreadsheet format. P2
FR-ANL-05	Analytics figures are computed in advance rather than on each request, so dashboards respond immediately. P2

5.16 Administration and audit ADM

FR-ADM-01	An administrator may create, edit and deactivate staff accounts. P1
FR-ADM-02	Role assignments are granted and revoked with an explicit scope, and a person may hold several at once. P1
FR-ADM-03	Grades, classes, subjects and terms are maintained by an administrator. P1
FR-ADM-04	Every creation, modification and deletion of a record is written to an audit log with actor, timestamp, entity and change. P1
FR-ADM-05	The principal may search the audit log by actor, entity or date range. P2

6 Screens and wireframes

The application presents a role-aware home screen above five fixed tabs. Announcements sit behind a control in the header on every screen rather than occupying a tab.

6.1 Screen inventory

#	Screen	Available to
1	Splash and session check	All
2	Sign in	All
3	Set or reset password	All
4	Home — role aware	All
5	Student search and directory	All but staff
6	Class list	Principal, sectional head
7	Class detail and roster	Principal, sectional head, class teacher
8	Student profile — overview	Principal, sectional head, class teacher, administrator
9	Student profile — attendance	As above
10	Student profile — marks	Principal, sectional head, class teacher
11	Student profile — activity	As screen 8
12	Student profile — benefits	Principal, administrator, own class teacher
13	Mark attendance	Class teacher, cover teacher
14	Attendance submitted	Class teacher, cover teacher
15	Record early leave	Class teacher, cover teacher
16	Attendance board — students	Principal, sectional head
17	Attendance board — staff	Principal, sectional head
18	Mark entry	Subject teacher
19	Marks review	Principal, sectional head
20	Staff directory	All
21	Staff profile	All, contents by

permission

22	Duty roster	All
23	My leave	All
24	Leave requests — pending list	Principal
25	Leave request detail	Principal
26	Assign cover teacher	Principal, sectional head
27	Announcement feed	All
28	Compose announcement	Principal, sectional head, administrator
29	Announcement detail	All
30	Inventory list	Principal, administrator
31	Inventory item detail and movements	Principal, administrator
32	Event calendar	All
33	Event detail and editor	Principal, administrator
34	School diary timeline	All
35	Diary entry detail and editor	Principal, administrator
36	Academic calendar setup	Principal
37	Calendar day editor	Principal
38	Attendance analytics	Principal, sectional head
39	Academic performance	Principal, sectional head
40	Reports and exports	Principal
41	User accounts and roles	Principal, administrator
42	Classes, subjects and terms	Principal, administrator
43	Audit log	Principal
44	My profile and settings	All
45	More menu	All, filtered by permission

6.2 Navigation

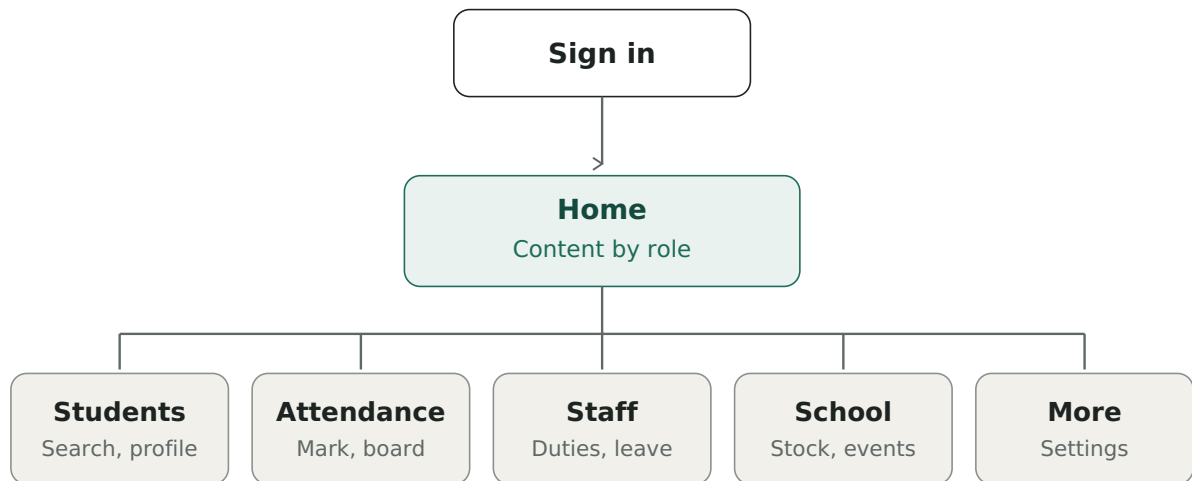


Figure 2. Top-level navigation. The same five tabs are present for every role; their contents are filtered by permission.

6.3 Wireframes — the daily attendance loop

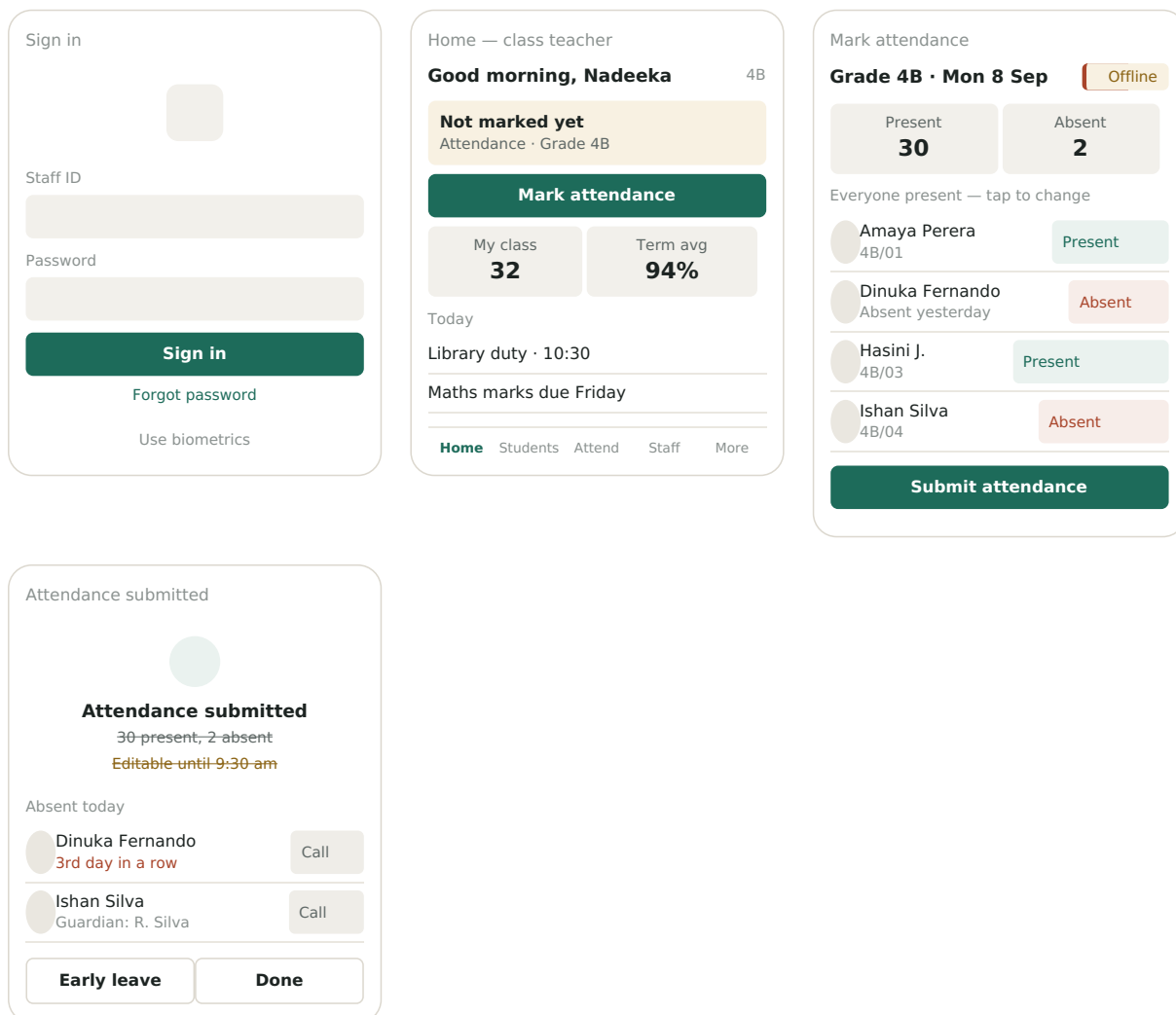


Figure 3. The class teacher's daily path: four taps from lock screen to submitted, with follow-up on absentees offered at the point of submission.

6.4 Wireframes — oversight

Home — principal

Good morning, Principal8:42

4 classes not marked
1C · 3A · 4D · 5B

Send reminder

Attendance
94.2%

Staff present
33/35

Absent today
52

Approvals
3

Needs attention

3 leave requests

5 students at absence risk

2 items low on stock

HomeStudentsAttendStaffMore

Attendance board — students

Mon 8 SepAll classes

Marked
26

Pending
4

Grade 1

1A1B1C1D1E

Grade 2

2A2B2C2D2E

Grade 3

3A3B3C3D3E

Remind 4 pending classes

Attendance board — staff

Mon 8 Sep35 staff

Present
31

Leave
2

Missing
2

Not checked in

S. Rajapaksa
3A · class unmarked

Call

M. Bandara
Music · no class today

Call

On approved leave

N. Peiris
1C · covered by K. Silva

T. Anandan
4D · no cover

Assign

Leave request detail

T. Anandan4D

Casual · 10–11 Sep
2 days · applied 7 Sep

Leave balance

Casual3 of 7 used

Medical1 of 14 used

Impact

4D needs a cover teacher for 2 days

2 other staff already on leave

Assign cover

RejectApprove

Figure 4. Oversight screens. Each leads with what requires action rather than with a summary — unmarked classes before percentages, staff not checked in before staff present, coverage impact before the approve button.

6.5 Wireframes — records and setup

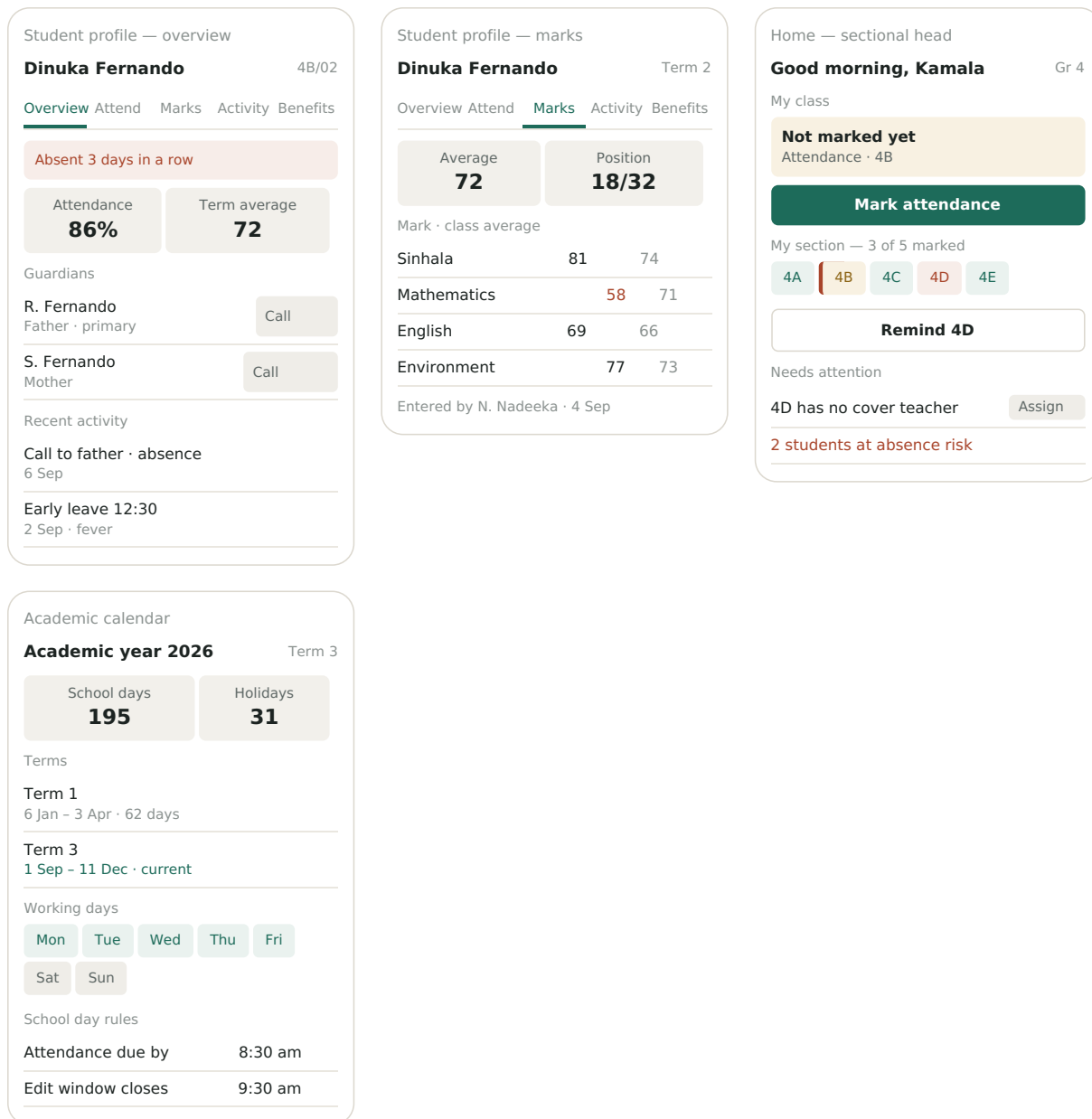


Figure 5. Record and configuration screens. The academic calendar governs every scheduled behaviour in the system and must be configured before attendance begins.

7 Data model

Roughly thirty tables. The set below is the intended structure; column-level definitions are given in the companion build specification.

7.1 Entities

Entity	Holds
academic_years	Year label, start and end dates, current flag
terms	Term within a year, start and end dates
calendar_days	Day-level overrides only: holiday, half day, exam, closure, with label

school_settings	Working-day pattern, school hours, attendance deadline, edit window, risk thresholds
grades	Grades 1 to 5
classes	Class within a grade, with its class teacher and academic year
subjects	Subject, and the grades it applies to
class_subject_teachers	Which teacher teaches which subject to which class
students	Admission number, name, date of birth, photograph, status
student_enrolments	A student's class for a given academic year — preserves history across promotion
guardians	Name, relationship, contact numbers
student_guardians	Links students to guardians; carries the primary-contact flag
staff	Staff number, name, contact number, joining date, status
roles	Named roles
permissions	Permission strings
role_permissions	Which permissions each role grants
staff_roles	A role held by a staff member at a scope — school, grade or class
cover_assignments	Temporary class teacher rights for a date range
student_attendance	One row per student per school day: status, marked by, marked at
attendance_submissions	One row per class per school day: submitted by, submitted at, locked at
early_leaves	Student, date, time, reason, collected by, recorded by
staff_attendance	Staff member, date, check-in time, status
leave_types	Casual, medical, duty, half day, and their annual entitlements
leave_balances	Entitlement and consumption per staff member, type and year
leave_requests	Request, dates, reason, attachment, status, approver, remarks
mark_sheets	Class, subject, term, status of draft, submitted or reopened
marks	Student, mark sheet, score, entered by, entered at
achievements	Student, title, category, level, date
memberships	Student, group, position, years held
benefits	Student, scheme, period, status, issued by
responsibilities	Staff member, duty or committee, position, academic year
announcements	Title, body, priority, audience, publish time, author
announcement_reads	Who has read which announcement, and when
inventory_items	Name, category, location, condition, minimum quantity
inventory_transactions	Receipt, issue, return or write-off, with quantity and actor
events	Title, dates, category, location, responsible staff, reminder lead times
diary_entries	Date, title, body, author, photographs, linked event
call_logs	Student or staff member called, caller, purpose, timestamp

notifications	Recipient, type, payload, sent and read timestamps
devices	Push tokens per staff member and device
audit_log	Actor, action, entity, entity identifier, change, timestamp
attendance_summaries	Pre-computed attendance figures by student, class, grade and period

7.2 Rules the model must enforce

- A class has at most one attendance submission per school day.
- A student has at most one attendance row per date.
- Exactly one guardian per student carries the primary-contact flag.
- Terms within an academic year may not overlap.
- Approved leave for overlapping dates may not exist twice for one staff member.
- Inventory quantity is derived from transactions, never written directly.
- A student's class is held by enrolment year, so promotion does not overwrite history.

Why guardians matter in phase 1

The parent application is phase 2, but collecting and verifying 900 guardian contact numbers is the slowest part of that phase. The guardian tables are therefore built and populated in phase 1, and every logged call improves their accuracy before the parent app is written.

8 Non-functional requirements

8.1 Performance PRF

NFR-PRF-01	The home screen presents usable content within two seconds of opening the app on a mid-range Android device over a 3G connection. P1
NFR-PRF-02	A class roster of forty students loads within one second. P1
NFR-PRF-03	Marking a student present or absent registers immediately in the interface, independent of network state. P1
NFR-PRF-04	Any dashboard or analytics screen renders within two seconds, achieved by reading pre-computed summaries rather than aggregating on demand. P2
NFR-PRF-05	The system supports at least forty concurrent users during the morning marking period without degradation. P1

8.2 Offline behaviour and reliability REL

NFR-REL-01	Attendance marking, early-leave recording and mark entry function fully without connectivity. P1
------------	---

NFR-REL-02	Unsyncronised entries survive the app being closed, the device restarting, and up to seven days offline. P1
NFR-REL-03	The synchronisation state of any locally held record is visible to the user at all times. P1
NFR-REL-04	Resubmitting the same attendance record does not create a duplicate. P1
NFR-REL-05	Where two users have modified the same record, the later server-side write prevails and the conflict is recorded in the audit log. P2

8.3 Security SEC

NFR-SEC-01	All traffic between app and server uses TLS. No school data is transmitted unencrypted. P1
NFR-SEC-02	Passwords are stored only as salted hashes. P1
NFR-SEC-03	Every permission check is performed on the server. The client may hide controls, but hiding a control is never the mechanism of enforcement. P1
NFR-SEC-04	Database access is constrained by row-level policies so that a compromised client cannot read records outside its scope. P1
NFR-SEC-05	Repeated failed sign-in attempts are rate limited. P1
NFR-SEC-06	Locally cached data is removed from the device on sign-out and on account deactivation. P1
NFR-SEC-07	Uploaded files are served only through authenticated, time-limited links. P2

8.4 Privacy and data protection PRV

NFR-PRV-01	The system holds personal data about children. Every field collected must have a stated purpose, and fields without one are not collected. P1
NFR-PRV-02	Guardian contact numbers are visible only to users with a working need — the child's own teachers, sectional head, principal and administrator. P1
NFR-PRV-03	Benefit records, which disclose family circumstances, are restricted further still, and are excluded from every export not explicitly authorised by the principal. P1
NFR-PRV-04	Student photographs are stored only where the school holds consent, and consent status is recorded. P1
NFR-PRV-05	Records of students who have left are retained according to a stated retention period and then archived or removed. P2
NFR-PRV-06	No school data is sent to any third-party analytics or advertising service. P1

8.5 Usability USE

NFR-USE-01	A class teacher can complete attendance for a class of forty students in under thirty seconds. P1
NFR-USE-02	Every screen leads with what requires action. Summary figures follow; they never displace an outstanding task. P1
NFR-USE-03	A class teacher's home screen offers exactly one obvious action before the attendance deadline. P1
NFR-USE-04	Every interactive control is at least 44 by 44 device-independent pixels. P1
NFR-USE-05	Destructive and irreversible actions require confirmation and state their consequence. P1
NFR-USE-06	A member of staff can perform their daily tasks without training beyond a single briefing. P1
NFR-USE-07	Error messages state what happened and what to do next, in plain language. P1

8.6 Accessibility ACC

NFR-ACC-01	Text meets a contrast ratio of at least 4.5:1 against its background. P1
NFR-ACC-02	Status is never conveyed by colour alone; a label or icon accompanies every colour-coded state. P1
NFR-ACC-03	The interface remains usable when the device font size is increased to its largest standard setting. P2

8.7 Compatibility CMP

NFR-CMP-01	The app supports Android 9 and above, and iOS 15 and above. P1
NFR-CMP-02	The app is usable on screens as narrow as 360 device-independent pixels. P1
NFR-CMP-03	Total download size stays below 40 MB. P2

8.8 Availability and continuity AVL

NFR-AVL-01	The service is available at least 99.5% of the time during school hours. P1
NFR-AVL-02	The database is backed up daily, and restoration to any point within the previous seven days is possible. P1
NFR-AVL-03	A restore is tested at least once before launch and once each year thereafter. P1

NFR-AVL-04	The school can export its complete data set in an open format at any time, so that it is never locked to one supplier. P1
------------	--

8.9 Localisation LOC

NFR-LOC-01	All interface text is held in resource files rather than embedded in code, so a further language can be added without changing screens. P1
NFR-LOC-02	Names and free text accept the full Unicode range, including Sinhala and Tamil script. P1
NFR-LOC-03	Dates and times are presented in the school's local timezone, and stored with timezone information. P1

8.10 Maintainability MNT

NFR-MNT-01	Roles, permissions, leave types, subjects, benefit schemes and inventory categories are configuration held as data, not values written into code. P1
NFR-MNT-02	Database schema changes are applied through versioned migrations held in source control. P1
NFR-MNT-03	Permission logic, attendance rules and calendar derivation are covered by automated tests. P1
NFR-MNT-04	The system is documented sufficiently for a developer other than the original author to make a change. P2

8.11 Scalability SCL

NFR-SCL-01	The design accommodates 1,500 students and 60 staff without structural change. P2
NFR-SCL-02	Ten years of attendance history — roughly two million rows — remains queryable within the stated performance limits. P2
NFR-SCL-03	The addition of the parent application in phase 2 requires no change to the data model described here. P1

9 Delivery phases

Stage	Contents	Proves
1.0	Data model, spreadsheet import, authentication, roles and scopes, academic calendar, student and staff directories	The foundation is correct
1.1	Student attendance, early leave, staff check-in, attendance board, unmarked-class reminder, absence-risk alerts	Teachers will use it daily
1.2	Marks entry and review, achievements, memberships, benefits, student timeline	The student record is complete
1.3	Announcements, leave requests and approval, cover assignment, responsibilities	Staff administration moves off paper
1.4	Inventory, events, school diary, analytics, exports, audit log	The school's records are in one place
2.0	Parent application	Out of scope for this document

Sequencing

Stages 1.0 and 1.1 carry the whole project. If teachers adopt daily attendance marking, every later stage lands on a system people already open each morning. If they do not, no amount of additional functionality will rescue it.

10 Assumptions and exclusions

10.1 Assumptions

- Every member of staff has a personal smartphone capable of running the app.
- The school has internet connectivity, though it may be unreliable inside classrooms.
- The school holds current contact numbers for staff, and for most guardians.
- Existing spreadsheet records are accurate enough to import after a reconciliation pass.
- One administrator will be responsible for account creation and data upkeep.
- The academic calendar will be configured before attendance marking begins.

10.2 Constraints

- Classroom connectivity is unreliable, which makes offline attendance marking mandatory rather than optional.
- Teachers have a narrow window before the school day begins; the marking flow must fit inside it.
- Staff have a wide range of technical confidence, which rules out interfaces requiring training.

10.3 Out of scope for this release

- The parent application, and any parent-facing notification.
- Fee collection, payments and accounting.
- Timetabling and period-level scheduling.
- Examination paper management and certificate generation.

- Library circulation and transport management.
- Biometric or card-based attendance hardware.
- Reporting to external education authorities.
- A web console. The system is mobile-only in this release.

11 Open decisions

Each of the following changes either the data model or the permission model and should be settled before the corresponding stage begins.

#	Decision	Needed by
1	How staff attendance is captured: self check-in, automatic on arrival at school, or a register kept by an administrator.	Stage 1.1
2	Whether sectional heads approve leave for their own section first, with the principal seeing only escalations and longer absences.	Stage 1.3
3	Whether a sectional head may see leave balances for teachers in their section.	Stage 1.3
4	Whether an administrator may edit the academic calendar, or the principal alone.	Stage 1.0
5	What a class teacher may see of a student in another class.	Stage 1.0
6	The thresholds that trigger an absence-risk alert: consecutive days, and percentage.	Stage 1.1
7	Whether parents will eventually see marks, or only attendance and announcements.	Stage 1.2
8	Whether a second language is required at launch or later.	Stage 1.0
9	Whether the app is distributed through the public app stores or privately to staff devices.	Stage 1.1

End of specification. The companion document, **AI Build Specification**, restates these requirements in the structured form needed to build the system.